Outline

- **Credit Hour: 3 Hours**
- **Duration: 3 Hours/Week**
- **Methods of student's assessment**
- **Continuous assessment 20**
- **Attendance and performance 10**
- **Assignments 10**
- **Written examination 60**

Contents

- Introduction.
- Basic Concepts and Measurement.
- Basic Quantities Vs Derived Quantities.
- Units of Measurement.
- Newton's Low of Motion.
- Waves and The Weight: The gravitational Force of the Earth .
- Work and Energy.
- Fluids Mechanics.

Contents

- Fluid at Rest: Density and Specific Gravity.
- Fluids at Rest: Pressure and Pressure Gauge.
- Medical Applications: Measurement of Human Blood Pressure.
- Wave Motion.
- Characteristic of Wave Motin.
- Type of Wave.
- Energy Transmitted by Wave Sound Wave
- Revision.

Physics and Measurement

Introduction

- **What is physics and Medical Physics?**
  - **Physics**

Physics is a branch of science aimed at describing the **Fundamental aspects** of our universe, this includes what things are in it, what properties of those things are noticeable, and what processes those things or their properties undergo. In simpler terms, physics attempts to **describe the basic mechanisms** that make our universe behave the way it does, and **to Discover the basic laws** by which it operates.

- **Physics Type**
  - **Theoretical** physics concentrates on developing the theory and mathematics of these laws, while
  - **Applied** physics focuses attention on the application of the principles of physics to practical problems.
  - **Experimental** physics lies at the intersection of physics and engineering, it has the theoretical knowledge of theoretical physicists, and they know how to build and work with scientific equipment.

- **Physics Division**

The major fields of physics are:

- Classical mechanics: is the study the motion of bodies according to Newton's laws of motion.
- Electricity and magnetism: are two closely related phenomena that are together considered a single field of physics.
- Quantum mechanics: describes the peculiar motion of very small bodies (atomic sizes and smaller).
- Optics: is the study of light.
- Acoustics: is the study of sound.
- Thermodynamics and statistical mechanics: are closely related fields that study the nature of heat.
- Solid-state physics: is the study of solids—most often crystalline metals.

- Plasma physics is the study of plasmas (ionized gases).
- **Atomic**, nuclear, and particle physics: study of the structure of the atom and the particles that make up the atom, its energy states and interaction, the atomic **nucleus** which study of physical and chemical properties of matter.
- Relativity: includes Albert Einstein's theories of special and general relativity. Special relativity describes the motion of bodies moving.
- Geophysics: is the study of the substances that make up the Earth and the physical processes occurring on, in and above it.
- Astronomy: is the study of the universe, the celestial objects like sun moon, planets and stars that make up the universe. The study has aided the setting up of communication satellites and the development of devices such as Global Positioning System (GPS)

Importance of Physics

- Physics provides the basic foundation on which other sciences are built. Which enables us to understand the basic components of matter and their mutual interactions.
- Knowledge and skills learnt in Physics find application in many areas of our daily lives. For example, proper use of household appliances such as refrigerators, iron boxes, television sets, replacing blown-up electric bulbs or fuses all require the knowledge learnt in Physics.
- Knowledge in Physics helps in the acquisition of career of well paying jobs. The most obvious careers are in the field of engineering which include: civil, electrical, mechanical, agricultural, chemical, computer engineering etc. Other careers are in the fields like, meteorology, computer science, laboratory, technology, surveying, geology and astronomy. Teaching, also useful to doctors, nurses and other science based careers.
- Technical instruments and equipment like meteorologists make use of instruments such as thermometers, barometers. In medical laboratories; electric microwaves are used in carrying out research on HIV (Human Immunodeficiency Virus) and (Acquired immunodeficiency Syndrome) AIDS and other diseases.

- **Medical physics**
  - Medical physics is a multidisciplinary field that combines the principles of physics with medical sciences to enhance healthcare through advanced diagnostic and therapeutic techniques.
  - This branch of science plays a pivotal role in the development and application of technologies such as X-rays, MRI, CT scans, and radiation therapy, all of which are essential tools in modern medicine. By understanding the fundamental principles of physics, medical physicists contribute to improving the accuracy and effectiveness of these technologies, ultimately leading to better patient outcomes.

Units of measurements (UOM)

- Physics is a quantitative science, we need to measure a number of quantities, e.g., size of objects, volume of liquids, amount of matter, weight of vegetables or fruits, body temperature, length of cloth, etc.
- In order to connect the physical world to the mathematical world, we need to make measurements of then real world. In making a measurement, we compare a physical quantity with some agreed-upon standard measuring unit which is internationally accepted, and determine how many such standard units are present. e.g, we have a precise definition of a unit of length called a mile, and have determined that there are about 92,000,000 such miles between the Earth and the Sun. For measuring the mass we need standard mass units of 1 kg, 500 g, etc. These standards are called **units**.
- The measured quantity is expressed in terms of a number followed by a corresponding unit, e.g., the Different quantities are measured in different units, e.g. length in meter (m), time in seconds (s), mass in kilogram (kg), etc. The standard measure of any quantity is called the unit of that quantity.

System of Units

- MKS: Meter Kilogram Second system (**SI Units System**).

Meter for Measuring Length, Kilogram for Measuring Mass, Second for Measuring Time

- CGS: Centimeter Gram Second system (**French Units System**).

Centimeter for Measuring Length, Gram for Measuring Mass, Second for Measuring Time

- FPS: Foot Pound Second system (**British Units System**).

Foot for Measuring Length, Pound for Measuring Mass, Second for Measuring Time

Physical quantities

- **Quantities**: Are any observable property or process in nature with which a number may be associated. The number may be obtained **directly** by a single measurement or **indirectly** e.g. by multiplying together two numbers obtained in separate operations of measurement.
- **Type of Quantities in terms of Origin and Formation**

  1. **fundamental quantities**: The physical quantities which do not depend on any other physical quantities for their measurements. There are seven fundamental quantities: length, mass, time, temperature, electric current, luminous intensity and amount of substance.
  2. **Fundamental Units**

Fundamental quantity	SI units	Symbol
1) Length	metre	m
2) Mass	kilogram	kg
3) Time	second	s
4) Temperature	kelvin	K
5) Electric current	ampere	A
6) Luminous Intensity	candela	cd
7) Amount of substance	mole	mol

2. Derived Quantities :

Which depend on **some or all** of the seven fundamental quantities. e.g area, volume, velocity, acceleration, density, weight, force, speed, momentum, resistance, conductivity, etc

- **Derived Units:**

Quantities	Derived Low	Derived Units
Area	length (m) * length (m).	m^2
Volume	length (m) * length (m) * length (m).	m3
Density	mass (kg) / volume (m^3)	kg/m^3
Velocity	displacement (*m*) / time taken (*s*)	m/s
Acceleration	change in velocity (*m/s*) / time taken (*s*)	m/s^2
Momentum	Mass*Velocity	kg m/s = kg m s^{-1}

3. Supplementary units :

Besides, the seven fundamental or basic units, there are two more units called supplementary units:

(i) Plane angle (dθ) : This is the ratio of the length of an arc of a circle to the radius of the circle as shown in Fig. 1.1 (a).

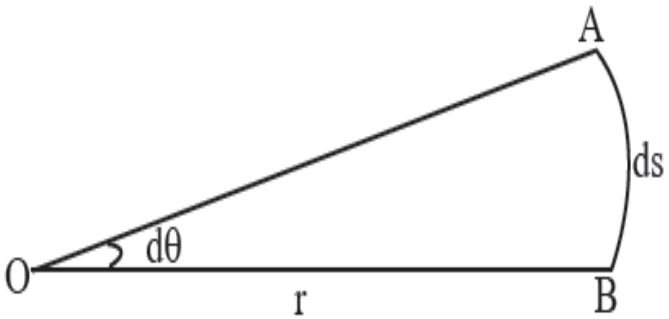

Fig 1.1 (a): Plane angle dθ.

dθ = ds/r

It is measured in radian (rad). An angle (θ) in radian is denoted as (θ^{o}) .

- **Supplementary units :**

(ii) Solid angle (dΩ) :

This is the 3-dimensional analogue of (dθ) and is defined as the area of a portion of surface of a sphere to the square of radius of the sphere.

Thus

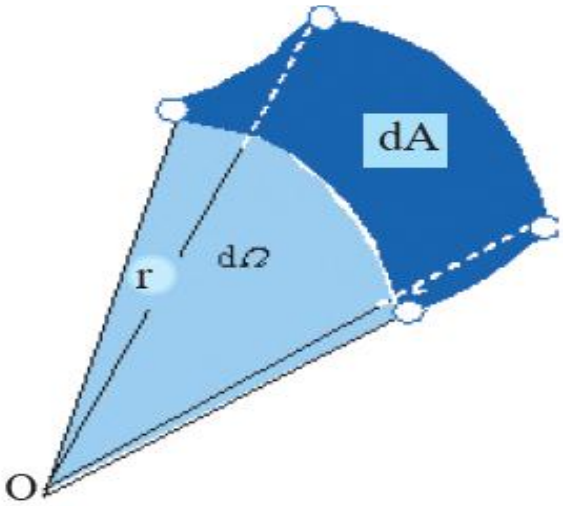

Fig 1.1 (b): Solid angle dΩ.

$$d\Omega = dA/r^2$$

is the solid angle subtended by area ds at O as shown in Fig. 1.1 (b).

It is measured in steradians (sr). A sphere of radius r has surface area ($4\pi r^2$). Thus, the solid angle subtended by the entire sphere at its center is:

$$\Omega = 4\pi r^2 / r^2 = 4\pi \text{ sr.}$$

- **Supplementary units (Cont.) :**

The relation between radian and degree is

π radians = π^{O} = 180°

1 radian = $\frac{180^{O}}{3.1415}$ = 57.297^{O}

Similarly $1^{O} = \frac{\pi}{180^{O}} = \frac{3.1415}{180} = 1.745*10^{-2}$ rad

1^{O} = 60 ´, 1 ´= $2.91*10^{-4}$ rad and

1 ´= 60 ˝, 1 ˝ = $4.4847*10^{-6}$ rad

Note: ((^{O}) ≡ arc degree, (´)≡ arcminute (˝) ≡ arcsecond)

Physical quantities (Cont.)

- **Type of Quantities in terms of Nature of the quantity:**

  1. **Scalar quantity**: a quantity that has magnitude (or size) only, but no direction. The magnitude is made up of a number and often an appropriate unit. Specific examples of scalar quantities are a distance of 2.5 m, a time interval of 15 s, a mass of 2.2 kg, and the grade of a mountain highway of 0.11 or 11%.

  2. **Vector quantity**: Is one that has both magnitude and direction. In this text, a vector quantity is indicated by a symbol with an arrow above it and the direction is stated in square brackets [ ] after the unit. A common vector quantity is position, which is the distance and direction of an object from a reference point. e.g, the position of a friend in your class could be at a distance of 2.2 m and in the west direction relative to your desk. The symbol for this position is $\vec{d}$ **= 2.2** m [W]. Another vector quantity is displacement, which is the change in position of an object in a given direction. The symbol for displacement is $\Delta\vec{d}$, where indicates change.

Prefixes for SI units

Prefix	Symbol	Magnitude
nano	n	10^{-9}
micro	μ	10^{-6}
mill	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
deca	da	10^{1}
hecto	h	10^{2}
kilo	k	10^{3}
Mega	M	10^{6}
Giga	G	10^{9}